Name : Roll No.: Class : Subject :

Assignment-7

Aim:
To redesign the interface of the project developed in the previous lab assignments for users with special needs and to create an accessible and inclusive interactive prototype.

Problem Statement:
In previous assignments you designed an application for the general population and created its wireframes, screens and prototype in Figma. In this assignment, use the same project and same figma file and redesign it for a group of users with special needs (for eg. users with visual impairment, hearing impairment, motor impairment). Identify the needs of the chosen user group, modify your existing screens and prototype to meet those needs by applying accessibility guidelines and check your final design with the accessibility checklist.

Theory:
Special needs design (also called accessible or inclusive design) means designing a product so that people with diff. abilities and limitations can use it effectively, efficiently and with satisfaction. A user may have permanent limitation (for example, blindness) or temporary (for eg. broken arm) or a

situational one (for eg. bright sunlight on the screen or a noisy place). A good interface for special needs is usually easier to use for every user not only for the target group.

Types of special needs considered while designing an interface:

→ Visual: blindness, low vision, colour blindness. Users depend on screen readers, magnification and strong contrast.

→ Hearing: deafness or hard of hearing. Users need captions, transcripts and visual alerts in place of audio only information.

→ Motor: limited hand movement, tremors, one handed use. Users need large touch targets, simple gestures and keyboard or voice operation.

→ Age related: elderly users

FAQ's:

1. Explain the 4 principles of WCAG.

A: Perceivable: means information should be easy to see or hear.
Operable: means the interface should be easy to navigate.
Understandable: means the content should be simple and clear.
Robust: should support assistive technologies eg. screen reader

2. Discuss design considerations for diff. impairments & how your interface supports them.

A. My website uses high contrast and clear text for visual impairment, for motor their are large buttons and hearing has visual alerts and written instructions.